Cover-Parameterised Hybrid Steganographic Model

$\mathcal{S}_{\mathrm{CPM}} = \underbrace{\mathrm{Synth}_{\mathrm{CP}}}_{\text{message-independent cover-parameter generation}} \triangleright \underbrace{\mathrm{Mask}}_{\text{keyed cover-conditioned payload protection}} \triangleright \underbrace{\mathrm{Embed}_{\mathrm{CM}}}_{\text{secret-bearing cover modification}}$

Shared secret key

$k \leftarrow \mathrm{Setup}(1^{\lambda})$

Public / Reproducible Context

Message length ℓ

Epoch context $\eta = \mathrm{Ctx}(\mathscr{F}_e)$

(Optional) metadata σ

Data Flow (public or internal)

Key/control dependency

Public channel / visible to warden

k

k

SENDER

Cover-parameter generation and keyed cover-conditioned masking

1 $\mathrm{Synth}_{\mathrm{CP}}(k, \ell)$

$(\gamma_1, \gamma_2) \leftarrow \mathrm{Synth}(k, \ell)$

$P_{\mathrm{params}} = G_1(\gamma_1) \oplus G_2(\gamma_2)$

$|P_{\mathrm{params}}| = \ell$

$\eta = \mathrm{Ctx}(\mathscr{F}_e)$

$d = \mathrm{Stream}_{K_{\mathrm{mask}}}(\eta)$

2 Parameter-conditioned mask

$\mu = P_{\mathrm{params}} \oplus d$

$|\mu| = \ell$

Secret message

$m \in \{0,1\}^{\ell}$

Cover object

$o \in O$

Secret bearing cover modification

3 $F_{mask}(m, P_{params})$

$b = m \oplus \mu$

$b \in \{0,1\}^{\ell}$

4 $\mathrm{Enc}_{\mathrm{CM}}(k, o, b)$

Adaptive embedding

$s \in S$

(stego-object)

Public Channel (Observable)

γ_1, γ_2, s

MAC_e, η

Adversary $\mathcal{A}$

$\mathrm{View}_{\mathcal{A}}$

$(\mathscr{F}_e, \eta, \ell, \sigma)$

Key and plaintext are not provided

RECEIVER

Verify full-epoch MAC before extraction

5 $\mathrm{Dec}_{\mathrm{CM}}(k, s)$

$\hat{b} = \mathrm{Dec}_{\mathrm{CM}}(k, s)$

$\hat{b} \in \{0,1\}^{\ell}$

Reproduce mask μ

$P_{\mathrm{params}} = G_1(\gamma_1) \oplus G_2(\gamma_2)$

$d = \mathrm{Stream}_{K_{\mathrm{mask}}}(\eta)$

$\mu = P_{\mathrm{params}} \oplus d$

6 Unmask

$F_{\mathrm{unmask}}(\hat{b}, P_{\mathrm{params}}; \eta)$

$m' = \hat{b} \oplus \mu$

$m' \in \{0,1\}^{\ell}$

Recovered message

$m' = m$